

Entanglement, Black Holes and the $ER = EPR$ Conjecture

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19 May 2026

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résumé.

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0. Introduction

The relationship between quantum mechanics and gravity remains one of the central open problems in theoretical physics. Currently, we have two theories that describe their respective domains with remarkable success, yet all attempts to reconcile them have run into difficulties. This tension becomes especially clear in the context of black holes, where seemingly simple assumptions lead to deep conceptual contradictions. Historically, however, such contradictions have often led to progress and driven major advances in our understanding of nature, including modern ideas suggesting that spacetime itself may emerge from quantum entanglement [1].

One example is the black hole information paradox, originating from Hawking's discovery that black holes emit thermal radiation and may delete information in the process [2,3]. The paradox was later made precise by Almheiri, Marolf, Polchinski, and Sully in the now famed AMPS paradox, which highlights the crucial role of quantum entanglement [4].

This thesis attempts to motivate and introduce the $\text{ER} = \text{EPR}$ conjecture proposed by Susskind and Maldacena as a possible resolution to the AMPS paradox [5]. The conjecture can be summarised as the proposed equivalence between entanglement (EPR) and Einstein-Rosen bridges (ER) [6,7], offering a concrete realisation of the broader idea that spacetime may be connected to quantum entanglement.

We proceed in roughly three parts. We discuss the EPR and ER aspects of $\text{ER} = \text{EPR}$ in chapters 1 and 2 respectively, before turning to the information paradox and $\text{ER} = \text{EPR}$ in chapters 3 and 4.

For simplicity, and due to matters of scope, we minimise the use of the AdS/CFT correspondence. However, since it plays an important role in motivating $\text{ER} = \text{EPR}$, we briefly introduce the correspondence and consider an example in this context. Throughout, we use natural units with $c = \hbar = k_B = 1$, while keeping G explicit.

1. Quantum Entanglement

Quantum mechanics usually concerns quantum states $|\Psi\rangle$ and their dynamics. However, such a framework is less than ideal when considering mixtures of states, or ensembles, and composite systems. This chapter introduces the nicer density matrix formalism which is well suited to handle these, especially when subsystems may be inaccessible to an observer.¹ We also discuss several constructions which will play an important role later, including the thermofield double and the Rindler decomposition.

1.1. Quantum ensembles

The meaning of an “ensemble” is best illustrated with an example. Consider a Stern-Gerlach-like setup in which an oven emits spin- $\frac{1}{2}$ particles with randomly oriented spins. Before any measurement is performed, the beam is described by the mixture or ensemble

$$\text{beam} \sim \{50\% : |\Psi_+\rangle, \quad 50\% : |\Psi_-\rangle\}$$

which cannot in general be represented by a single state $|\Psi_{\text{SG}}\rangle$. [8]

More generally, consider an ensemble of states $|\Psi_i\rangle$, where the system is prepared in the state $|\Psi_i\rangle$ with probability p_i . We impose the p_i sum to unity

$$\sum_i p_i = 1 \tag{1.1}$$

as probabilities should. The expectation value of an operator $\mathcal{O}$ is then given by [8–10]

$$\langle \mathcal{O} \rangle = \sum_i p_i \langle \Psi_i | \mathcal{O} | \Psi_i \rangle \tag{1.2}$$

Now, define the density matrix [8–10]

$$\rho \equiv \sum_i p_i |\Psi_i\rangle \langle \Psi_i| \tag{1.3}$$

which allows us to write the expectation value (1.2) as

$$\langle \mathcal{O} \rangle = \text{tr}[\rho \mathcal{O}]$$

implying the density matrix contains all measurable information about the system.

Also, the normalisation condition (1.1) becomes

$$\text{tr} \rho = 1 \tag{1.4}$$

¹i.e. the interior of a black hole.

We call the state pure if the density matrix can be written as

$$\rho = |\Psi\rangle\langle\Psi|$$

implying $\rho^2 = \rho$ and $\text{tr} \rho^2 = 1$. Otherwise, the state is called mixed.

1.1.1. Unitary evolution

Consider evolving the system with a unitary operator $U(t)$. The states $|\Psi_i\rangle$ evolve according to [8]

$$|\Psi_i(t)\rangle = U(t)|\Psi_i(0)\rangle$$

while the probabilities p_i remain unchanged. The density matrix therefore evolves as

$$\begin{aligned} \rho(t) &= \sum_i p_i |\Psi_i(t)\rangle\langle\Psi_i(t)| \\ &= U(t)\rho(0)U^\dagger(t) \end{aligned} \tag{1.5}$$

Now, using $U^\dagger(t)U(t) = \mathbb{1}$ and the cyclicity of the trace we obtain from Equation (1.5)

$$\text{tr}[\rho(t)^2] = \text{tr}[\rho(0)^2]$$

Hence, a pure state cannot become mixed under unitary evolution.

1.2. The entangled state

Consider a system with multiple degrees of freedom, such as the string of qubits shown in Figure 1.

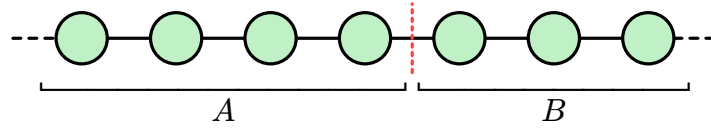


Figure 1: A string of qubits.

Suppose the total system is in a pure state $|\Psi\rangle$ with density matrix

$$\rho_{AB} = |\Psi\rangle\langle\Psi| \tag{1.6}$$

We imagine dividing the total system into two subsystems A and B . This allows us to decompose the Hilbert space as

$$\mathcal{H}_{AB} = \mathcal{H}_A \otimes \mathcal{H}_B$$

Now, consider two observers: Alice and Bob. Alice only has access to A , while Bob only has access to B . This means Alice can perform measurements on A , but has no access

to the degrees of freedom in B . To Alice the system is therefore fully described by the reduced density matrix [8–10]

$$\rho_A \equiv \text{tr}_B \rho_{AB} \quad (1.7)$$

where tr_B denotes the partial trace over B . Applying tr_B removes the degrees of freedom inaccessible to Alice. To see why the definition (1.7) is natural consider any operator corresponding to a measurement Alice can perform. Such an operator acts only on states in $\mathcal{H}_A$ and therefore takes the form

$$\mathcal{O} = \mathcal{O}_A \otimes \mathbb{1}_B$$

Using (1.2) we then obtain [8–10]

$$\begin{aligned} \langle \mathcal{O} \rangle &= \text{tr}_{AB} [\rho_{AB} (\mathcal{O}_A \otimes \mathbb{1}_B)] \\ &= \text{tr}_A [\text{tr}_B (\rho_{AB}) \mathcal{O}_A] \\ &= \text{tr}_A [\rho_A \mathcal{O}_A] \end{aligned}$$

Thus the result of any measurement Alice can perform is completely determined by the reduced density matrix.

The structure of the reduced density matrix depends on the form of the total state $|\Psi\rangle$. We call the state $|\Psi\rangle$ separable if it can be written as a product

$$|\Psi\rangle = |\Psi\rangle_A \otimes |\Psi\rangle_B$$

implying

$$\rho_{AB} = \rho_A \otimes \rho_B$$

where both ρ_A and ρ_B are pure. Conversely, if the total state $|\Psi\rangle$ is pure but not separable, the reduced density matrices

$$\rho_A = \text{tr}_B \rho_{AB}, \quad \rho_B = \text{tr}_A \rho_{AB}$$

are mixed. Such states are called entangled.

1.2.1. The EPR pair I

The simplest entangled states are the EPR pairs.² For a system of two qubits with bases $\{|0\rangle_A, |1\rangle_A\}$ and $\{|0\rangle_B, |1\rangle_B\}$ an example of such a pair is [8–10]

²or Bell states.

$$|\text{EPR}\rangle = \frac{1}{\sqrt{2}}(|0\rangle_A \otimes |1\rangle_B + |1\rangle_A \otimes |0\rangle_B) \equiv \frac{1}{\sqrt{2}}(|01\rangle + |10\rangle) \quad (1.8)$$

Now, imagine Alice performs a measurement on her qubit and obtains 0. The state (1.8) then collapses as

$$|\text{EPR}\rangle \xrightarrow{\text{Alice m.m.}} |01\rangle$$

implying that Bob's qubit must be in the state $|1\rangle$. Likewise, if Alice measures 1, Bob's qubit must be in the state $|0\rangle$. Their measurement outcomes are therefore perfectly anti-correlated.

These correlations persist even when Alice and Bob are separated by arbitrarily large distances. This seems highly non-local and was the basis of the Einstein-Podolsky-Rosen paradox [6]. However, no violation of locality occurs, since the correlations only become observable once Alice and Bob compare their measurements.

1.3. Entanglement entropy

We would like a number quantifying the mixedness of a state. This number is the von Neumann entropy [8–10]

$$S \equiv -\text{tr}[\rho \ln \rho] \quad (1.9)$$

The density matrix can always be diagonalised³ in which case the von Neumann entropy becomes

$$S = -\sum_{i=1}^N \rho_{ii} \ln \rho_{ii} \quad (1.10)$$

where ρ_{ii} are the eigenvalues of ρ .

To see the definition (1.9) is sensible, consider:

1. A pure state. In this case

$$\rho = |\Psi_j\rangle\langle\Psi_j|$$

so

$$\rho_{jj} = 1, \quad \rho_{ii} = 0$$

for $i \neq j$. The entropy (1.10) therefore vanishes

$$S = -\ln 1 = 0$$

³This follows from $\rho^\dagger = \rho$. Also, the trace is independent of our choice of basis.

2. A completely mixed state. Suppose we have N states $|\Psi_i\rangle$ occurring with equal probability $p_i = N^{-1}$. Then

$$\rho = \frac{1}{N} \mathbb{1}_N$$

or $\rho_{ii} = N^{-1}$. The entropy (1.10) is therefore

$$S = - \sum_{i=1}^N \frac{1}{N} \ln \frac{1}{N} = \ln N$$

We find the von Neumann entropy vanishes for pure states and is positive for mixed states, with the maximal value being $\ln N$, as expected for a measure of mixedness.

Now, for a pure state $|\Psi\rangle$, we define the entanglement entropy as the von Neumann entropy of the reduced density matrix [9,10]

$$S_{\text{EE}} \equiv S_A = -\text{tr}[\rho_A \ln \rho_A] \quad (1.11)$$

The entanglement entropy measures how entangled the state $|\Psi\rangle$ is. To see this, consider taking $|\Psi\rangle$ to be:

1. A separable state. In this case ρ_A is pure, implying

$$S_{\text{EE}} = 0$$

2. An entangled state. In this case ρ_A is mixed, implying

$$S_{\text{EE}} \neq 0$$

The difference occurs because we discard information stored in correlations between A and B when tracing out B , thereby increasing the entropy $S_{\text{EE}} \neq 0$.

1.3.1. The EPR pair II

As an example, consider again the EPR pair. The density matrix of (1.8) is

$$\rho_{AB} = \frac{1}{2} \{ |01\rangle\langle 01| + |01\rangle\langle 10| + |10\rangle\langle 01| + |10\rangle\langle 10| \}$$

and taking the partial trace over B gives

$$\rho_A = \frac{1}{2} |0\rangle\langle 0| + \frac{1}{2} |1\rangle\langle 1|$$

which is maximally mixed, implying $S_A = \ln 2$.

1.4. Thermal systems

With the essential formalism in place, we now turn to systems in which ensembles show up naturally: thermal systems, as is well known from statistical mechanics. [11]

Consider a system in thermal equilibrium described by the Hamiltonian H . Let $|E_i\rangle$ denote a complete set of energy eigenstates satisfying

$$H|E_i\rangle = E_i|E_i\rangle$$

Using the basis $\{|E_i\rangle\}$, the density matrix (1.3) becomes

$$\rho = \sum_i p_i |E_i\rangle \langle E_i|$$

where p_i is the probability that the system occupies the state $|E_i\rangle$.

For a canonical ensemble at temperature T , the probabilities are given by the Boltzmann distribution⁴ [11]

$$p_i \propto e^{-\beta E_i} \tag{1.12}$$

where $\beta = T^{-1}$. Substituting (1.12) into the density matrix gives

$$\rho \propto \sum_i e^{-\beta E_i} |E_i\rangle \langle E_i|$$

Since the $|E_i\rangle$ form an eigenbasis of H , we may write

$$\rho \propto e^{-\beta H}$$

Using the normalisation condition (1.4) then gives the thermal density matrix

$$\rho = \frac{e^{-\beta H}}{Z(\beta)}; \quad Z(\beta) = \text{tr } e^{-\beta H} = \sum_i e^{-\beta E_i} \tag{1.13}$$

where $Z(\beta)$ is the partition function. [8,12]

The von Neumann entropy of (1.13) is [12]

$$S = \beta \langle H \rangle + \ln Z \tag{1.14}$$

which coincides with the thermodynamic entropy [11]. To better understand the thermal density matrix, consider:

1. The high-temperature limit $\beta \rightarrow 0$. For a N -dimensional Hilbert space

$$\rho \rightarrow \frac{1}{N} \mathbb{1}_N$$

⁴A less handwavy justification is given in [12].

implying

$$S \rightarrow \ln N$$

2. The low-temperature limit $\beta \rightarrow \infty$. Assuming a non-degenerate ground state

$$\rho \rightarrow |E_0\rangle\langle E_0|$$

implying

$$S \rightarrow 0$$

So the thermal density matrix is mixed for all $T > 0$, becoming pure as $T \rightarrow 0$.

1.4.1. The thermofield double

Since the thermal density matrix is mixed, it should be possible to obtain it from a pure state living in a larger Hilbert space. The choice of purification is not unique, but a convenient one is found by introducing a second copy of the system. This leads to the thermofield double [13–16]

$$|\text{TFD}\rangle = \frac{1}{\sqrt{Z(\beta)}} \sum_i e^{-\beta E_i/2} |E_i\rangle_A \otimes |E_i\rangle_B \quad (1.15)$$

This state purifies the thermal density matrix. To see this we trace over B giving

$$\begin{aligned} \rho_A &= \text{tr}_B(|\text{TFD}\rangle\langle\text{TFD}|) \\ &= \frac{1}{Z(\beta)} \sum_i \sum_j e^{-\beta(E_i+E_j)/2} {}_A\langle E_i| \langle E_j|_A \underbrace{\sum_k {}_B\langle E_k|E_i\rangle_B \langle E_j|E_k\rangle_B}_{=\delta_{ij}} \\ &= \frac{1}{Z(\beta)} \sum_i e^{-\beta E_i} {}_A\langle E_i| \langle E_i|_A \end{aligned}$$

or

$$\rho_A = \frac{e^{-\beta H}}{Z(\beta)}$$

recovering (1.13).

Physically, the thermofield double describes an entangled pure state between two identical copies of a system. Since the reduced density matrix of either subsystem is thermal, an observer with access to only one copy sees a thermal state.

1.5. A detour

We take a brief detour to introduce a method for determining the ground state of a system using path integrals. The starting point is the time-evolution operator⁵

$$U(T) = e^{-iHT}$$

whose transition amplitudes can be written as a path integral [14,16,17]

$$\langle \phi_2 | e^{-iHT} | \phi_1 \rangle = \int_{\phi(t=0)=\phi_1}^{\phi(t=T)=\phi_2} D\phi e^{-S[\phi]}$$

with S being the Lorentzian action. The Euclidean path integral is given by a Wick rotation $t = -it_E$

$$\langle \phi_2 | e^{-\beta H} | \phi_1 \rangle = \int_{\phi(t_E=0)=\phi_1}^{\phi(t_E=\beta)=\phi_2} D\phi e^{-S_E[\phi]} \quad (1.16)$$

with S_E being the Euclidean action.

We think of $\phi_{1,2}$ as boundary conditions. The meaning of this depends on the space we integrate over. As an example for $\mathbb{R}^d$ we would integrate over the strip $\mathbb{R}^{d-1} \times I_\beta$ as shown for $d = 2$ in Figure 2. [16]

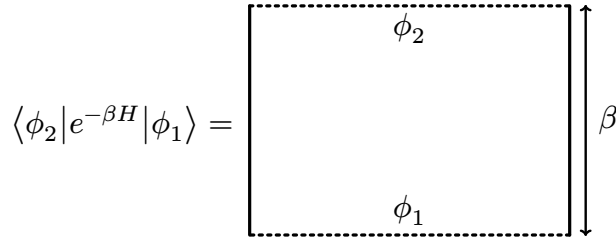


Figure 2: Example path integral for $\mathbb{R}^2$. [16]

While if space were a sphere S^d we would integrate over the cylinder $S^{d-1} \times I_\beta$.

So the path integral requires boundary data on the cuts $t_E = 0$ and $t_E = \beta$. We can think of (1.16) with the cut $t_E = \beta$ open as the state $|\Psi\rangle$

$$|\Psi\rangle = e^{-\beta H} |\phi_1\rangle$$

where we integrate over the geometry in Figure 3. [16]

⁵assuming H is time-independent.

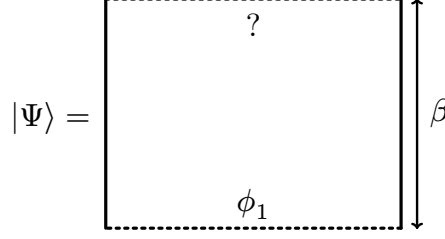


Figure 3: Example path integral for $\mathbb{R}^2$ with an open cut. [16]

Then (1.16) defines a wavefunctional of the boundary field $\Psi[\phi_2] = \langle \phi_2 | \Psi \rangle$. This allows us to read off the state $|\Psi\rangle$. Then any path integral with an open cut Σ defines or prepares a state on Σ . [16]

1.5.1. Determining the ground state

Now, we can define the ground state of a system by a path integral extending infinitely in t_E as in Figure 4. [16]

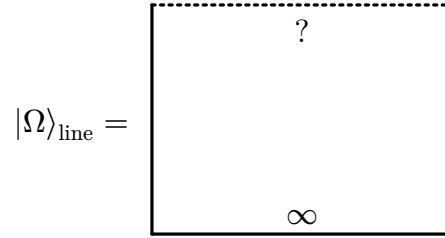


Figure 4: Example path integral to prepare the ground state.

To see this expand $|\Psi\rangle$ in the energy eigenstates of the Hamiltonian denoted by $|E_n\rangle$

$$|\Psi\rangle = \sum_n c_n |E_n\rangle$$

Evolving in t_E we obtain

$$\begin{aligned} |\Psi(t_E)\rangle &= \sum_n c_n e^{-E_n t_E} |E_n\rangle \\ &= e^{-E_0 t_E} \sum_n c_n e^{-(E_n - E_0) t_E} |E_n\rangle \end{aligned}$$

Now, taking $t_E \rightarrow \infty$ we find⁶

$$|\Psi(t_E)\rangle \propto |E_0\rangle$$

since $E_n - E_0 > 0$. So taking $t_E \rightarrow \infty$ projects onto the ground state.

⁶assuming $|E_0\rangle$ is non-degenerate.

1.6. The entangled vacuum

Consider a quantum field $\phi(x)$ in its vacuum state $|\Omega\rangle$ and divide space into two regions A and B . The two-point correlator is given by [15,17]

$$\langle\Omega|\phi(x)\phi(y)|\Omega\rangle \stackrel{\text{scalar field}}{=} \frac{1}{4\pi^2} \frac{m}{|x-y|} K_1(m|x-y|) \quad (1.17)$$

Take $|x-y| \ll m^{-1}$ then

$$K_1(m|x-y|) \sim \frac{1}{m|x-y|}$$

so (1.17) becomes

$$\langle\Omega|\phi(x)\phi(y)|\Omega\rangle \sim \frac{1}{|x-y|^2}$$

which diverges as $|x-y| \rightarrow 0$. Then the correlation between points x and y becomes formally infinite as we bring them together.⁷

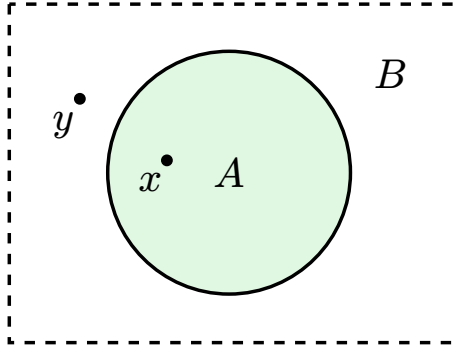


Figure 5: Example division of space into regions A and B .

We now take $x \in A$ and $y \in B$ as in Figure 5. Then (1.17) implies the regions A and B are entangled. To see this assume the vacuum is not entangled

$$|\Omega\rangle = |\Omega\rangle_A \otimes |\Omega\rangle_B$$

Then

$$\langle\Omega|\phi(x)\phi(y)|\Omega\rangle = {}_A\langle\Omega|\phi(x)|\Omega\rangle_{AB}\langle\Omega|\phi(y)|\Omega\rangle_B$$

and since $\langle\Omega|\phi(x)|\Omega\rangle = 0$ we find

$$\langle\Omega|\phi(x)\phi(y)|\Omega\rangle = 0 \sim \text{contradiction}$$

implying the vacuum is entangled.

⁷We have assumed ϕ is a scalar field, however, this behaviour is seen more generally.

1.7. The Rindler decomposition

To better understand the entanglement structure of the Minkowski vacuum we can determine the ground state $|\Psi\rangle_{\text{M}}$. Computing $|\Psi\rangle_{\text{M}}$ is also a nice warm-up before diving into black holes where we will do something completely analogous. We do this by considering the Rindler decomposition.

The idea is to pick a spatial coordinate, say x , and split the Hilbert space as [14,15,18]

$$\mathcal{H} = \mathcal{H}_L \otimes \mathcal{H}_R$$

with $\mathcal{H}_R$ being acted on by fields with $x > 0$ and $\mathcal{H}_L$ being acted on by fields with $x < 0$. We would like a simple basis for $\mathcal{H}_L$ and $\mathcal{H}_R$. To this end we consider the generator of Lorentz boosts. The metric in Minkowski spacetime is⁸

$$ds^2 = -dt^2 + dx^2 \tag{1.18}$$

and the boost generator K_x is given by

$$K_x = x\partial_t + t\partial_x$$

We want coordinates adapted to the flow of K_x such that using these coordinates K_x becomes the generator of time translations.

These coordinates are the Rindler coordinates (η, ξ) defined for $x > |t|$ by

$$t = \xi \sinh \eta; \quad x = \xi \cosh \eta \tag{1.19}$$

with $\xi > 0$. We call $x > |t|$ the right Rindler wedge.⁹ With (1.19) the metric (1.18) becomes

$$ds^2 = -\xi^2 d\eta^2 + d\xi^2 \tag{1.20}$$

which is the Rindler metric. The surfaces $x = \pm t$ define the Rindler horizons and correspond to $\xi = 0$ in the Rindler coordinates. A uniformly accelerated observer follows a trajectory of constant $\xi > 0$, satisfying

$$x^2 - t^2 = \xi^2$$

which corresponds to a hyperbola in Minkowski spacetime. Such observers remain in the right Rindler wedge and can never receive signals from behind the Rindler horizons. The right Rindler wedge is shown in Figure 6.

⁸we suppress $dy^2 + dz^2$.

⁹An analogous definition works for the left Rindler wedge $x < -|t|$.

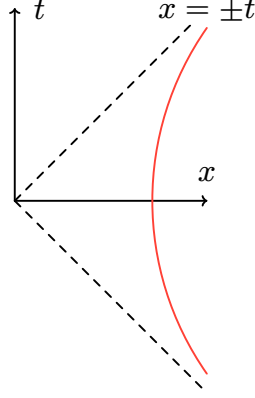


Figure 6: The right Rindler wedge in Minkowski coordinates (x, t) . The dashed lines ($x = \pm t$) are the Rindler horizons and the red hyperbola ($x^2 - t^2 = \xi^2$) is representative of the trajectory for a Rindler observer.

However, as opposed to other horizons these are a property of the observer and not the spacetime. This is obvious since the spacetime is Minkowski in disguise. We can think of these horizons arising due to the observer accelerating *away* from the spacetime beyond them.

With the Rindler coordinates (1.19) the boost generator becomes

$$\partial_\eta = x\partial_t + t\partial_x = K_x$$

so K_x acts as the Hamiltonian generating translations in the Rindler time η .

To compute the path integral and determine $|\Psi\rangle_M$ we need an appropriate Euclidean geometry. This is obtained by a Wick rotation $\eta \rightarrow -i\eta_E$, after which the Rindler coordinates (1.19) become

$$t_E = \xi \sin \eta_E; \quad x = \xi \cos \eta_E \quad (1.21)$$

and the metric (1.20) becomes

$$ds^2 = \xi^2 d\eta_E^2 + d\xi^2 \quad (1.22)$$

which is the polar metric

$$ds^2 = \rho^2 d\theta^2 + d\rho^2 \quad (1.23)$$

with the identification $\rho = \xi$ and $\theta = \eta_E$.

Since spacetime is still Minkowski nothing weird should happen so we require the metric (1.22) is regular or *smooth* at the horizon $\xi = 0$. This implies η_E is periodic

$$\eta_E \sim \eta_E + 2\pi$$

The Minkowski wavefunctional on the slice $t = 0$ is now defined by evaluating the path integral over $t_E \leq 0$. From (1.21) we see that K_x generates rotations in the xt_E -plane. Euclidean time evolution generated by K_x is then given by $e^{-\eta_E K_x}$, where η_E is an angle in the xt_E -plane. The region $t_E \leq 0$ therefore corresponds to evolution by an angle π , as shown in Figure 7.

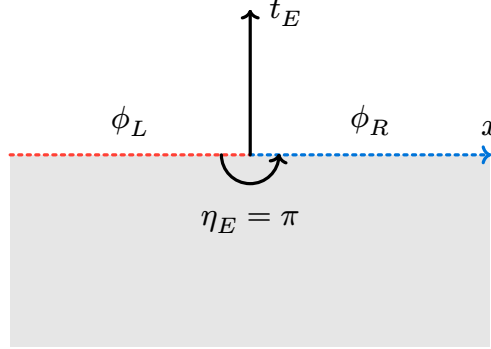


Figure 7: Geometry for defining the Minkowski vacuum wavefunctional using the Rindler decomposition. We use the lower-half plane as an initial condition.

With the above reasoning the path integral (1.16) can be written as¹⁰ [14–16]

$$\langle \phi_L \phi_R | \Omega \rangle \propto \langle \phi_R | e^{-\pi K_R} \Theta | \phi_L \rangle$$

where K_R is the restriction of K_x to the right Rindler wedge and Θ is the CPT operator. The role of Θ is to map states in $\mathcal{H}_L$ to corresponding states in $\mathcal{H}_R$ by $\Theta | \phi_L \rangle \in \mathcal{H}_R$, allowing the boost operator K_R to act on them. Now, we can use completeness of the K_R eigenstates $|i\rangle_R$ to write

$$\begin{aligned} \langle \phi_L \phi_R | \Omega \rangle &\propto \sum_i \langle \phi_R | e^{-\pi K_R} | i \rangle_R \langle i |_R \Theta | \phi_L \rangle \\ &\propto \sum_i e^{-\pi E_i} \langle i |_R \Theta | \phi_L \rangle \langle \phi_R | i \rangle_R \\ &\propto \sum_i e^{-\pi E_i} \langle \phi_L | i^* \rangle_L \langle \phi_R | i \rangle_R \end{aligned}$$

where $|i^*\rangle_L = \Theta^\dagger |i\rangle_R$. Then by inspection the Minkowski vacuum is

$$|\Psi\rangle_M = \frac{1}{\sqrt{Z}} \sum_i e^{-\pi E_i} |i^*\rangle_L \otimes |i\rangle_R \quad (1.24)$$

¹⁰We have $\phi_1 = \phi(\eta_E = 0) = \phi_R$ and $\phi_2 = \phi(\eta_E = \pi) = \phi_L$.

which we recognise as the thermofield double (1.15) with $\beta = 2\pi$. The entanglement structure is now completely manifest, with the left and right Rindler wedges clearly being entangled.

Note, the reduced density matrix of (1.24) is

$$\rho_R = \frac{1}{Z} \sum_i e^{-2\pi E_i} |i\rangle_R \langle i|_R$$

which is thermal with the Unruh temperature¹¹ [19]

$$T_{\text{Unruh}} = \frac{1}{2\pi}$$

This is the Unruh effect where accelerating observers experience a temperature. Which we now see is related to the left Rindler wedge being inaccessible to these observers.

¹¹With the omission of a length scale.

2. Black Holes

Black holes show up as solutions to the Einstein field equations and provide the setting for Einstein-Rosen bridges. This chapter introduces classical- and quantum black holes in Minkowski spacetime, including the Hartle-Hawking state and black hole thermodynamics. We also discuss black holes in anti-de Sitter spacetime, where an ER = EPR like connection appears.

2.1. The Schwarzschild metric

Consider the source-free Einstein field equations

$$R_{\mu\nu} = 0 \tag{2.1}$$

By the Birkhoff-Jebsen theorem, the unique spherically symmetric solution of (2.1) that asymptotically approaches Minkowski spacetime is the Schwarzschild metric. The Schwarzschild metric is given by [15,18,20,21]

$$ds^2 = -f(r) dt^2 + f(r)^{-1} dr^2 + r^2 d\Omega^2 \tag{2.2}$$

where

$$f(r) \equiv 1 - \frac{r_s}{r} = \frac{r - r_s}{r} \tag{2.3}$$

with $r_s \equiv 2GM$ being the Schwarzschild radius. The metric (2.2) describes the spacetime outside any spherically symmetric mass distribution of total mass M centered at $r = 0$.

We see that the metric components diverge at $r = 0$ and $r = r_s$. The point $r = 0$ is a true singularity, since curvature diverges as $r \rightarrow 0$. The divergence at the Schwarzschild radius, however, is caused by a bad choice of coordinates, and nothing special happens locally as $r \rightarrow r_s$.¹² We call the Schwarzschild radius *the horizon*.

The horizon is significant due to the sign of $f(r)$ flipping upon crossing r_s . Then continuing toward smaller r corresponds to moving forward in time. Any object entering $r < r_s$ is therefore destined to reach the singularity ($r = 0$) since doing otherwise corresponds to travelling backwards in time or running from the future. Thus crossing

¹²at least classically.

the horizon is a one-way trip. We can define black holes to be objects whose physical size is smaller than their corresponding Schwarzschild radius.

2.2. The fully-extended geometry

To better understand the geometry near the horizon, we seek coordinates in which the metric (2.2) is non-singular at $r = r_s$. We first define the tortoise coordinate r_* through [18,21]

$$dr_* = \frac{dr}{f(r)} \quad (2.4)$$

(2.4) can be integrated giving

$$r_* = r + r_s \log\left(\frac{r - r_s}{r_s}\right) \quad (2.5)$$

Using (2.5) we define the Kruskal-Szerkeres coordinates [18,21]

$$U \equiv -\exp\left(\frac{r_* - t}{2r_s}\right) \quad (2.6)$$

$$V \equiv \exp\left(\frac{r_* + t}{2r_s}\right) \quad (2.7)$$

which satisfy

$$UV = \frac{r_s - r}{r_s} e^{r/r_s}$$

With (2.6) the metric (2.2) becomes

$$ds^2 = -\frac{2r_s^3}{r} e^{-r/r_s} (dU dV + dV dU) + r^2 d\Omega^2 \quad (2.8)$$

where r is implicitly defined by $r \equiv r(U, V)$. We can introduce coordinates that make (2.8) diagonal by defining

$$T \equiv \frac{V + U}{2} \sim \text{time}$$

$$X \equiv \frac{V - U}{2} \sim \text{space}$$

Then (2.8) takes the form

$$ds^2 = \frac{4r_s^3}{r} e^{-r/r_s} (-dT^2 + dX^2) + r^2 d\Omega^2 \quad (2.9)$$

We see (2.9) is non-singular at the horizon, while also being conformally Minkowski since

$$ds^2 \propto -dT^2 + dX^2$$

implying null radial geodesics satisfy

$$dT = \pm dX$$

The geometry described by (2.9) is shown in Figure 8

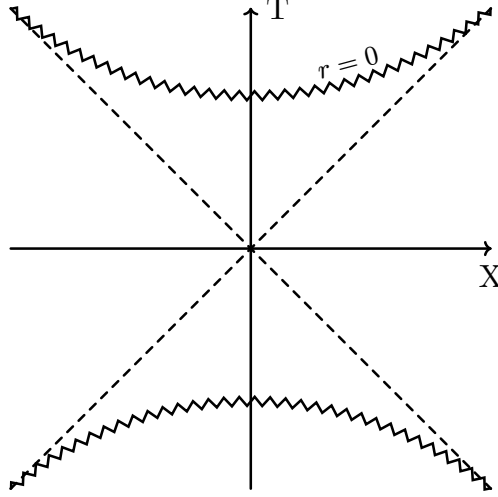


Figure 8: The Kruskal extension. The dashed lines correspond to the horizon where $U = 0$ or $V = 0$, implying $T = \pm X$, while the jagged lines correspond to the singularity.

2.3. Penrose diagrams and Einstein-Rosen bridges

We are often interested in the causal structure of spacetime. This motivates the construction of Penrose diagrams, which compactify spacetime by mapping infinities to finite distances in a conformal way. Starting with (2.8) we can make the coordinates finite by applying the transformation [18]

$$\tilde{U} = \arctan U; \quad \tilde{V} = \arctan V$$

with $\tilde{U}, \tilde{V} \in (-\pi/2, \pi/2)$. Then (2.8) becomes¹³

$$ds^2 = \Omega^{-2}(\tilde{U}, \tilde{V}) d\tilde{U} d\tilde{V} \tag{2.10}$$

where

$$\Omega^{-2}(\tilde{U}, \tilde{V}) = \frac{4r_s^3}{r} \frac{e^{-r/r_s}}{\cos^2 \tilde{U} \cos^2 \tilde{V}}$$

We again define

¹³We suppress $d\Omega^2$.

$$\tilde{T} \equiv \frac{\tilde{V} + \tilde{U}}{2} \sim \text{time}$$

$$\tilde{X} \equiv \frac{\tilde{V} - \tilde{U}}{2} \sim \text{space}$$

Then (2.10) takes the form

$$ds^2 \propto -d\tilde{T}^2 + d\tilde{X}^2$$

which has the same causal structure as (2.9).

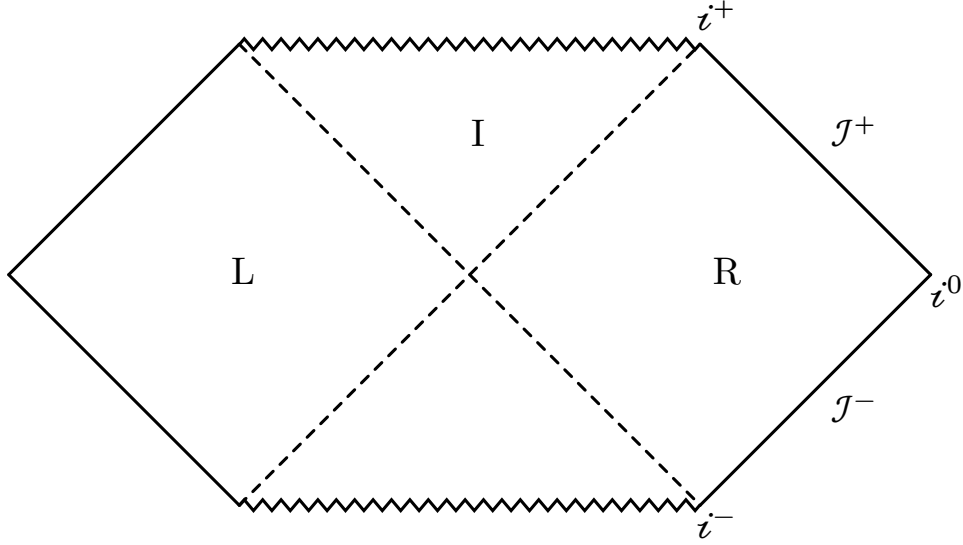


Figure 9: Penrose diagram for the fully-extended Schwarzschild geometry. The dashed lines correspond to the horizons ($r = r_s$) and the jagged lines correspond to the singularities ($r = 0$). $i^\pm$ are the past and future timelike infinities, i^0 is spatial infinity, and $J^\pm$ are the past and future null infinity. [15,18,21]

The Penrose diagram for the fully-extended Schwarzschild geometry is shown in Figure 9. [15,18,21] The spacetime contains two asymptotically Minkowski exterior regions, L and R , where $r \rightarrow \infty$, together with past and future singularities at $r = 0$. The two exteriors appear connected through an Einstein-Rosen bridge or wormhole [7]. However, the diagram makes clear that L and R are causally disconnected. An observer beginning in either L or R may cross the horizon and enter the black hole interior I . We know this is a one-way journey towards the singularity and the diagram makes this obvious.

Now, consider two observers, one originating in L and the other in R , each crossing their respective horizons. Although the exterior regions are causally disconnected, the observers may nevertheless meet inside the interior I before reaching the singularity. So the Einstein-Rosen bridge is non-traversable, yet the future interiors of L and R overlap causally.

L and R are often interpreted as distinct asymptotically Minkowski universes connected through the Einstein-Rosen bridge. Alternatively, one may attempt to interpret them as widely separated regions of the same universe, corresponding heuristically to two black holes separated by a large distance. With this interpretation the bridge appears highly non-local. Observers beginning arbitrarily far apart in the exterior geometry may encounter each other inside I . However, this interpretation is only approximate due to the mutual attraction between the black holes.

The fully-extended Schwarzschild geometry described above is highly idealised and does not correspond to real black holes formed by collapse. The Penrose diagram for such a black hole can be constructed by combining the Penrose diagrams for Minkowski space (see Figure 10a) and the extended Schwarzschild geometry as shown in Figure 10b. The resulting spacetime contains only a single interior region and does not include an Einstein-Rosen bridge or second asymptotic region. [15,18]

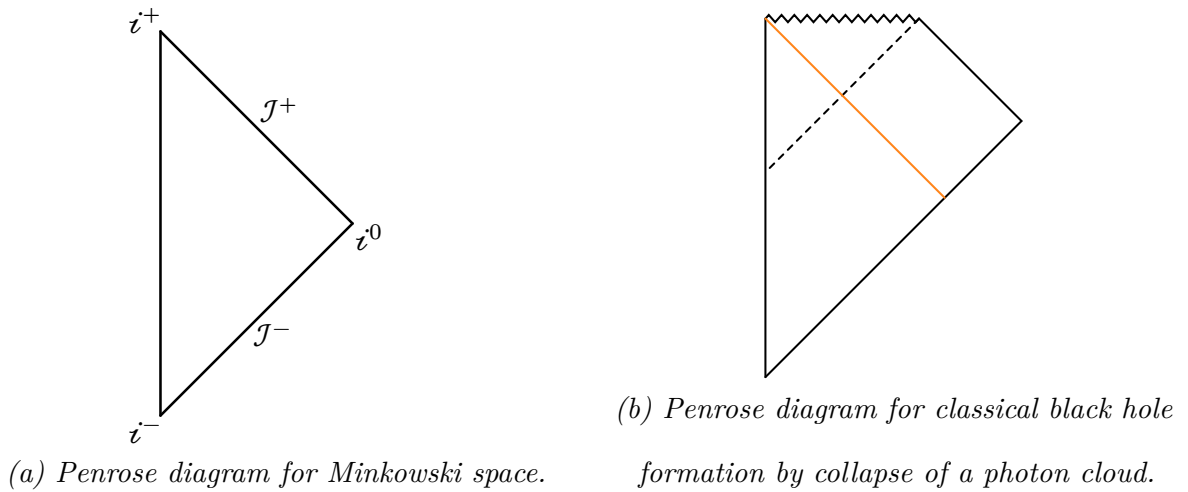


Figure 10: Penrose diagrams.

2.4. The Hartle-Hawking state

We would like to determine the ground state for the fully-extended Schwarzschild geometry. This is done by computing the path integral as when we determined $|\Psi\rangle_M$.

Consider the Schwarzschild metric (2.2)

$$ds^2 = -\frac{r-r_s}{r} dt^2 + \frac{r}{r-r_s} dr^2 + r^2 d\Omega^2$$

The appropriate Euclidean geometry is obtained after a Wick rotation $t = -it_E$

$$ds^2 = \frac{r-r_s}{r} dt_E^2 + \frac{r}{r-r_s} dr^2 + r^2 d\Omega^2 \quad (2.11)$$

We now write

$$r = r_s + \delta$$

with $0 < \delta \ll r_s$. This is called the near-horizon limit and will prove useful. Using $f(r_s) = 0$ we expand $f(r)$ around r_s as

$$f(r) \stackrel{\text{Taylor}}{\simeq} \underbrace{f(r_s)}_0 + \frac{df(r_s)}{dr} \delta \simeq \frac{\delta}{r_s}$$

Also, $r^2 \simeq r_s^2$ so (2.11) becomes¹⁴

$$ds^2 \simeq \frac{\delta}{r_s} dt_E^2 + \frac{r_s}{\delta} d\delta^2 \quad (2.12)$$

We can define $\rho^2 \equiv 4r_s \delta$ to write (2.12) as

$$ds^2 \simeq \left(\frac{\rho}{2r_s} \right)^2 dt_E^2 + d\rho^2 \quad (2.13)$$

(2.13) is the Rindler metric (1.22) and we say the horizon is locally Rindler. Like in the Minkowski case the metric (2.13) becomes the polar metric (1.23) upon defining θ by

$$t_E \equiv 2r_s \theta$$

We call the geometry described by (2.13) the Euclidean cigar. [14–16] This is shown in Figure 11.

¹⁴We suppress the 2-sphere.

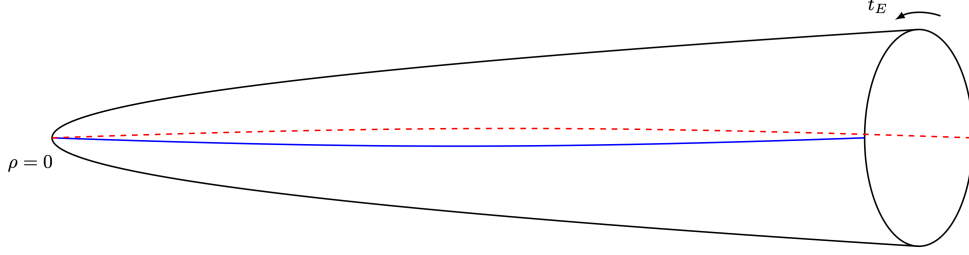


Figure 11: The Euclidean cigar. [15]

By the equivalence principle nothing special happens at the horizon. We therefore require the Euclidean cigar is smooth at $\rho = 0$. This implies $\theta \sim \theta + 2\pi$ or equivalently

$$t_E \sim t_E + \beta$$

with $\beta \equiv 4r_s \pi$.

The Hartle-Hawking wavefunctional [22,23] on the slice $t = 0$ is now defined by evaluating the path integral over the lower-half of the Euclidean cigar. This region corresponds to the interval

$$-\frac{\beta}{2} < t_E < 0$$

and the construction is shown in Figure 12. Euclidean time evolution generated by H is given by $e^{-t_E H}$, so evolution over the interval above is given by $e^{-\beta H/2}$.

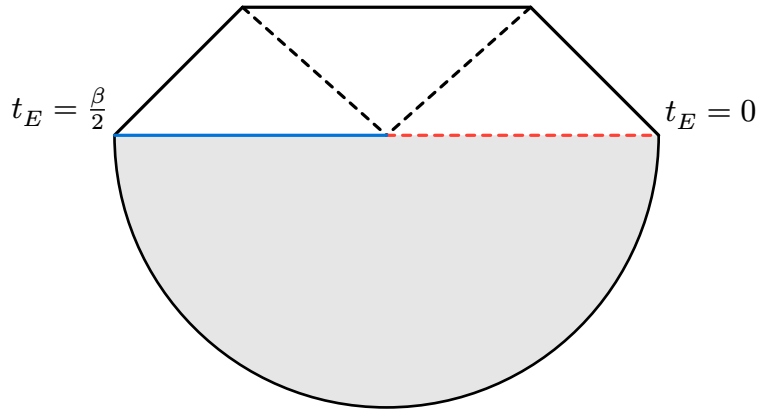


Figure 12: Geometry for defining the Hartle-Hawking wavefunctional. The upper-half is just the upper-half of the usual Penrose diagram. [15]

With the above reasoning the path integral (1.16) can be written as¹⁵ [14–16]

$$\langle \phi_L \phi_R | \Omega \rangle \propto \langle \phi_R | e^{-\beta H_R/2} \Theta | \phi_L \rangle$$

¹⁵We have $\phi_1 = \phi(t_E = 0) = \phi_R$ and $\phi_2 = \phi(t_E = \frac{\beta}{2}) = \phi_L$.

where H_R is the restriction of the Hamiltonian to R and Θ is the CPT-operator. As with the Minkowski vacuum we use completeness to find [15]

$$\langle \phi_L \phi_R | \Omega \rangle \propto \sum_i e^{-\beta E_i/2} \langle \phi_L | i^* \rangle_L \langle \phi_R | i \rangle_R$$

where $|i^*\rangle_L = \Theta^\dagger |i\rangle_R$. We can then read off the Hartle-Hawking state

$$|\Psi\rangle_{\text{HH}} = \frac{1}{\sqrt{Z(\beta)}} \sum_i e^{-\beta E_i/2} |i^*\rangle_L \otimes |i\rangle_R \quad (2.14)$$

which we recognise as the thermofield double (1.15). Therefore the regions L and R are quantum mechanically entangled, while geometrically they are connected by an Einstein-Rosen bridge. The fully-extended Schwarzschild geometry can, as noted, be viewed as corresponding to two black holes separated by a large distance. Then the Hartle-Hawking state tells us these will be entangled¹⁶.

We briefly note the Hartle-Hawking state was derived using two widely accepted assumptions:

1. Validity of the equivalence principle. This assumption is implicit upon requiring the horizon is smooth.
2. Validity of working with quantum fields on a fixed classical background. This assumption is implicit when computing the path integral on the Schwarzschild background.

2.5. Black holes as thermal systems

The reduced density matrix of the Hartle-Hawking state is thermal with the Hawking temperature [2,15,23]

$$T_H = \beta^{-1} = \frac{1}{4\pi r_s} \quad (2.15)$$

This means an observer confined to either region L or R sees a thermal state with temperature T_H because L and R are entangled.

The Hartle-Hawking state describes a black hole in equilibrium with its own radiation. [15] This equilibrium arises because we consider a spacetime which has contained

¹⁶in a heuristic manner.

a black hole for an infinite amount of time.¹⁷ The cause of the thermal bath can then only be the black hole with the system having plenty of time to reach equilibrium. This implies the black hole has temperature T_H and emits radiation. We could now imagine placing or forming a black hole in a cold universe. This black hole still has temperature T_H and will radiate. However, this black hole would not be in equilibrium and will begin evaporating.

Since a black hole has some temperature T_H and an energy which we can identify as $E = M$ it will have some entropy. By definition [11]

$$\frac{dS}{dE} = \frac{1}{T_H}$$

Using $S(E = 0) = 0$ we find

$$S = \frac{A}{4G} = 2\pi \frac{A}{\ell_p^2} \quad (2.16)$$

with $A = 4\pi r_s^2$ being the area of the horizon and $\ell_p = \sqrt{8\pi G}$. The quantity (2.16) is referred to as the Bekenstein-Hawking entropy. [2,24]

2.6. in anti-de Sitter

The Schwarzschild metric (2.2) asymptotically approaches Minkowski spacetime which has a vanishing cosmological constant. We now want a metric that asymptotically approaches anti-de Sitter spacetime. The metric for empty¹⁸ anti-de Sitter spacetime has the same static, spherically symmetric form as (2.2)

$$ds^2 = -f(r) dt^2 + f(r)^{-1} dr^2 + r^2 d\Omega^2 \quad (2.17)$$

with [21,25]

$$f(r) = 1 + \frac{r^2}{r_{\text{ads}}^2}; \quad r_{\text{ads}}^2 \equiv -\frac{3}{\Lambda} \quad (2.18)$$

where $\Lambda < 0$. We combine (2.3) and (2.18) to obtain the metric describing a Schwarzschild black hole in anti-de Sitter spacetime by using

¹⁷which is clearly non-physical.

¹⁸Containing only a negative cosmological constant.

$$f = 1 - \frac{r_s}{r} + \frac{r^2}{r_{\text{ads}}^2}; \quad r_{\text{ads}}^2 = -\frac{3}{\Lambda} \quad (2.19)$$

The form of (2.19) can be justified by considering:

1. For $r \ll r_{\text{ads}}$, the anti-de Sitter term becomes negligible and

$$f \simeq 1 - \frac{r_s}{r}$$

recovering the Schwarzschild metric (2.3).

2. For $r \rightarrow \infty$,

$$f \simeq 1 + \frac{r^2}{r_{\text{ads}}^2}$$

recovering the anti-de Sitter metric (2.18).

So (2.19) reproduces the Schwarzschild geometry at small r and approaches anti-de Sitter spacetime as $r \rightarrow \infty$. This is exactly the behavior we wanted. The Penrose diagram describing (2.19) is seen in Figure 13. [15]

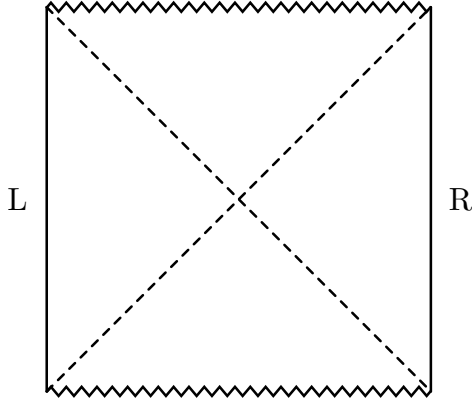


Figure 13: Penrose diagram for an eternal black hole in anti-de Sitter spacetime. Here, L and R are two timelike boundaries. [15]

The state describing this geometry is again the Hartle-Hawking state (2.14) [15,26].

We can determine the temperature of the black hole by defining a horizon radius r_+ with $f(r_+) = 0$ and expanding $f(r)$ as¹⁹

$$f(r) \stackrel{\text{Taylor}}{\simeq} \underbrace{f(r_+)}_0 + \left. \frac{df(r)}{dr} \right|_{r=r_+} \delta \quad (2.20)$$

with $\delta = r - r_+$ and

¹⁹Now, the Schwarzschild radius r_s is just a constant related to the mass M .

$$f(r_+) = 1 - \frac{r_s}{r_+} + \frac{r_+^2}{r_{\text{ads}}^2} \stackrel{!}{=} 0 \quad (2.21)$$

Then (2.20) becomes

$$f(r) \simeq \left(\frac{r_s}{r_+^2} + \frac{2r_+}{r_{\text{ads}}^2} \right) \delta \stackrel{\text{by (2.21)}}{=} \left(\frac{r_{\text{ads}}^2 + 3r_+^2}{r_+ r_{\text{ads}}^2} \right) \delta$$

The Euclidean metric after a Wick rotation $t = -it_E$ is then

$$ds^2 = \left(\frac{r_{\text{ads}}^2 + 3r_+^2}{r_+ r_{\text{ads}}^2} \right) \delta dt_E^2 + \left(\frac{r_+ r_{\text{ads}}^2}{r_{\text{ads}}^2 + 3r_+^2} \right) \frac{d\delta^2}{\delta} \quad (2.22)$$

We can define

$$\rho^2 \equiv \frac{4r_+ r_{\text{ads}}^2}{r_{\text{ads}}^2 + 3r_+^2} \delta$$

and write (2.22) as

$$ds^2 = \left(\frac{r_{\text{ads}}^2 + 3r_+^2}{2r_+ r_{\text{ads}}^2} \right)^2 \rho^2 dt_E^2 + d\rho^2 \quad (2.23)$$

(2.23) is again the Rindler metric (1.22) and becomes the polar metric (1.23) if we define θ by

$$t_E \equiv \frac{2r_+ r_{\text{ads}}^2}{r_{\text{ads}}^2 + 3r_+^2} \theta$$

The horizon is smooth when $\theta \sim \theta + 2\pi$ implying

$$t_E \sim t_E + \beta$$

with

$$\beta \equiv 4\pi \frac{r_+ r_{\text{ads}}^2}{r_{\text{ads}}^2 + 3r_+^2}$$

The Hawking temperature for a Schwarzschild black hole in anti-de Sitter spacetime is then [14]

$$T_H = \beta^{-1} = \frac{1}{4\pi} \frac{r_{\text{ads}}^2 + 3r_+^2}{r_+ r_{\text{ads}}^2} = \frac{1}{4\pi} \left(\frac{1}{r_+} + \frac{3r_+}{r_{\text{ads}}^2} \right) \quad (2.24)$$

with r_+ defined implicitly by $f(r_+) = 0$.

Now, consider:

1. When $r_+ \ll r_{\text{ads}}$ we recover $r_+ \simeq r_s$ from (2.21) and (2.24) becomes

$$T_H \simeq \frac{1}{4\pi r_s}$$

as we had before (2.15).

2. When $r_+ \gg r_{\text{ads}}$ we have from (2.21)

$$r_+ - r_s + \frac{r_+^3}{r_{\text{ads}}^2} \stackrel{!}{=} 0 \Rightarrow r_+ \simeq (r_s r_{\text{ads}}^2)^{1/3}$$

so (2.24) becomes

$$T_H \simeq \frac{1}{4\pi} \frac{3r_+}{r_{\text{ads}}^2} = \frac{3}{4\pi} \left(\frac{r_s}{r_{\text{ads}}^4} \right)^{1/3} \quad (2.25)$$

which increases linearly with the size of the black hole.

These are referred to as small and large black holes respectively. (2.25) implies large black holes are stable and behave *normally*²⁰, whereas small black holes are unstable. One interesting feature of anti-de Sitter spacetime is the two timelike boundaries L and R in Figure 13. Radiation can reach these boundaries in a finite time before being reflected and returning, so the anti-de Sitter boundary effectively acts like a reflecting box. Large black holes can therefore reach stable equilibrium with their own radiation, while small black holes remain unstable and continue to evaporate. Then eternal black holes in anti-de Sitter spacetime are meaningful, as opposed to eternal black holes in Minkowski spacetime, which are unphysical. [15,26]

2.7. The AdS/CFT correspondence

But, why do we care about anti-de Sitter spacetime? Our Universe appears to have a positive cosmological constant and would then be asymptotically de Sitter. Therefore, the apparent interest in anti-de Sitter seems unmotivated. However, we care because of the AdS/CFT correspondence due to Maldacena [27].

²⁰i.e they have positive heat capacities.

²¹The boundary of AdS is always of lower dimensionality as compared to the bulk. An example of the correspondence could be AdS³/CFT². The correspondence is therefore sometimes referred to as being a *holographic duality*.

The correspondence essentially states that a *bulk* theory of quantum gravity in anti-de Sitter spacetime is equivalent to a *boundary* conformal field theory.²¹ [15,28] This idea is illustrated in Figure 14.

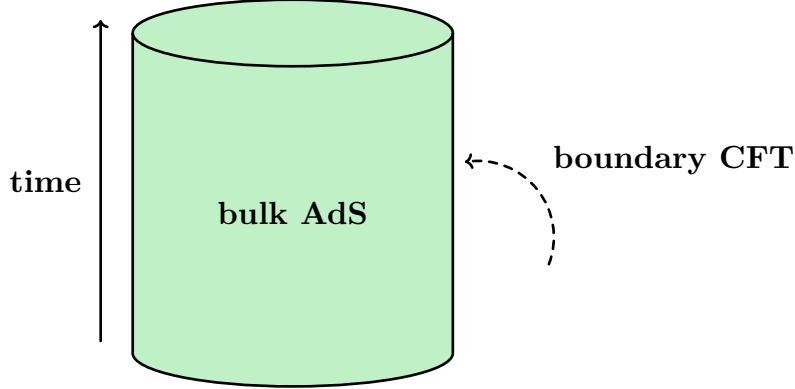


Figure 14: The AdS/CFT correspondence. A theory of gravity in the bulk AdS is equivalent to a CFT on the boundary. The boundary of AdS^3 is a cylinder $S_1 \times \mathbb{R}$ when compactified hence the figure.

Since CFTs are generally much better understood than theories of quantum gravity, this duality provides a powerful framework for studying quantum gravity. A full treatment of the correspondence is way beyond the scope of this project, but its key feature is that it provides an exact definition of the bulk theory in terms of the boundary CFT.

Previously, we interpreted the Hartle-Hawking state (2.14) as describing entanglement between the two bulk exteriors L and R , which are connected geometrically by an Einstein-Rosen bridge. The AdS/CFT correspondence refines this picture for black holes in anti-de Sitter spacetime. Now, each asymptotic region is dual to a separate boundary CFT, and the Hartle-Hawking state can more fundamentally be understood as an entangled state of these two CFTs. [15,16,26]

The Hartle-Hawking state then corresponds to a thermofield double (1.15) of two independent CFTs which is dual to a single connected spacetime [16,26]. The Einstein-Rosen bridge can thus be interpreted as the geometric realisation of the entanglement in the boundary theory. Symbolically we can write:

$$\text{ER bridge} \overset{\text{AdS/CFT}}{\Leftrightarrow} \text{TFD entanglement}$$

which is the most important $\text{ER} = \text{EPR}$ -like equivalence. We can consider this to be the canonical example of $\text{ER} = \text{EPR}$ [5].

3. The Information Paradox

Black holes being thermal with a temperature T_H leads to a tension between two otherwise sacred principles: unitarity and the equivalence principle. This chapter introduces this tension through the black hole information paradox and its entanglement-based formulation in the AMPS paradox.

3.1. Black hole evaporation

Black holes formed by collapse emit radiation and slowly lose mass. This process reduces the horizon area, so the black hole shrinks over time. The Penrose diagram for a black hole formed by collapse is seen in Figure 15.

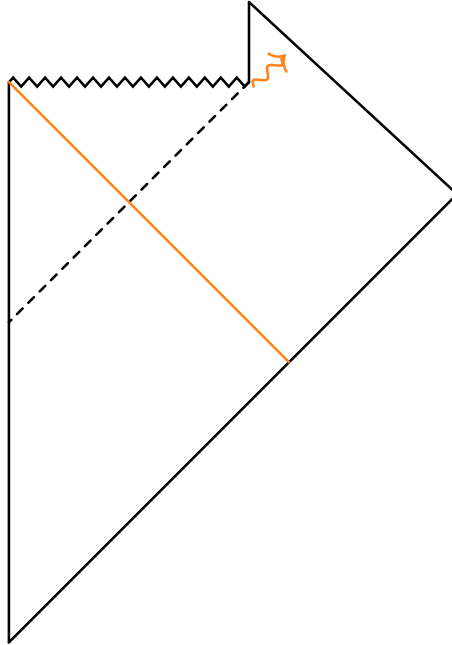


Figure 15: Penrose diagram for an evaporating black hole. [15]

This evaporation was shown to be problematic by Hawking [3,15,29–32]. To see why, consider a universe containing a black hole formed by collapsing a cloud of photons. When the black hole is formed and before it has had time to emit radiation the total state

$$|\Psi\rangle = |\Psi\rangle_{\text{BH}}$$

is pure. During evaporation, we assume the total Hilbert space may be decomposed as

$$\mathcal{H}_{\text{total}} = \mathcal{H}_{\text{BH}} \otimes \mathcal{H}_{\text{R}}$$

where $\mathcal{H}_{\text{BH}}$ describes interior degrees of freedom and $\mathcal{H}_{\text{R}}$ the emitted radiation. The total state is still pure

$$\rho_{\text{total}} = |\Psi\rangle\langle\Psi|$$

However, an exterior observer has no access to the internal degrees of freedom $\mathcal{H}_{\text{BH}}$ so their system is described by the reduced density matrix

$$\rho_{\text{R}} = \text{tr}_{\text{BH}} \rho_{\text{total}}$$

which is thermal.²² This can be quantified by saying

$$S_{\text{R}} > 0$$

and as the black hole evaporates ρ_{R} becomes increasingly mixed. The natural implication is that this proceeds until the black hole itself evaporates completely leaving us with a thermal state.²³ Then the total state has evolved from a pure state $|\Psi\rangle_{\text{BH}}$ to a mixed state ρ_{R} violating unitarity. This leads to information loss since we lose information encoded in the matter used to form the black hole when it is emitted as radiation. This is why we refer to this problem as the information paradox.²⁴

3.1.1. The Page curve

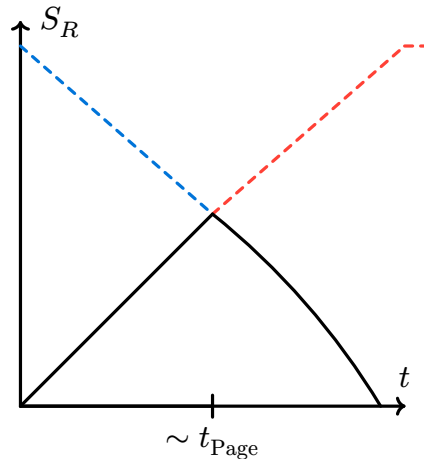


Figure 16: An example of what the Page curve could look like. The red dotted line represents non-unitary evaporation, while the black line represents unitary evaporation. The blue dotted line corresponds to the black hole entropy. [15,30,33]

²²assuming the horizon is smooth.

²³ignoring remnants.

²⁴For the sake of brevity we ignore objections to Hawking's argument, since the AMPS paradox is more robust.

The evolution of S_R is called the Page curve [15,33]. We trivially have $S_R = 0$ initially and as evaporation proceeds $S_R > 0$ as mentioned. When unitarity is violated S_R keeps increasing until the black hole is fully evaporated and $S_R \simeq S_{\text{BH}}^{\text{initial}}$. However, under the assumption that unitarity is sacred there should be some turnover point when $S_R \simeq S_{\text{BH}}$ such that $S_R \rightarrow 0$ eventually.

An example of what this curve could look like is shown in Figure 16. The details of the Page curve are not useful for our purposes. Note, however, that the final state being pure implies entanglement between late- and early radiation. This is needed to purify the late radiation and ensure $S_R \rightarrow 0$. We refer to the time this occurs as the Page time t_{Page} .

3.2. The AMPS paradox

Almheiri, Marolf, Polchinski, and Sully made the information paradox formulated by Hawking precise with the AMPS paradox [4]. We will consider a formulation of the AMPS paradox aligned with the original formulation illustrating the importance of the entanglement structure near the horizon.²⁵ Later, we consider a slightly different formulation in the context of ER = EPR where this is helpful.

Consider an old black hole evaporating, with old meaning we are past the Page time. We will assume the following: [4,32]

1. Validity of our semiclassical effective field theory description near the horizon.
2. The process is unitary.
3. The horizon is smooth.

The AMPS paradox shows these assumptions are mutually inconsistent, thus forming a *trilemma*. We further assume, motivated by the effective field theory description, that:

4. The Hilbert space can be decomposed as

$$\mathcal{H}_A \otimes \mathcal{H}_B \otimes \mathcal{H}_C \subset \mathcal{H} \tag{3.1}$$

²⁵originally posed to show BHC [34,35] is inconsistent.

where $\mathcal{H}_A$ describes degrees of freedom in the early radiation, $\mathcal{H}_B$ describes degrees of freedom in the late radiation, and $\mathcal{H}_C$ describes degrees of freedom inside the black hole.

This assumption seems reasonable, as such decompositions emerge naturally in local quantum field theory.²⁶ However, it is expected, if not known, that such a strict decomposition fails in theories of quantum gravity. This is not, by itself, an objection to the AMPS paradox, since the paradox is formulated within an effective regime where such a decomposition could still emerge. However, whether such an effective decomposition is always reliable in quantum gravity is unclear. [15,30–32,36] We touch more upon this later, and will find ER = EPR challenges this naive decomposition and thereby “avoids” the AMPS paradox in a clever way.

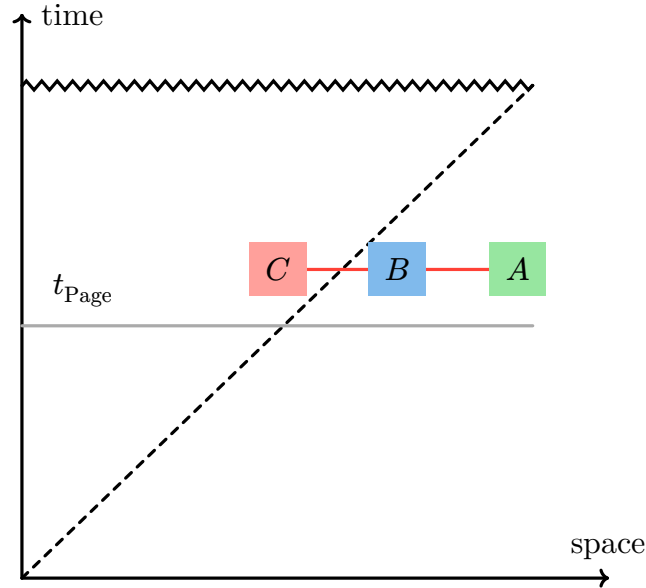


Figure 17: Sketch of the AMPS paradox. A is a mode of early radiation which has had time to propagate, B is a mode of late radiation sitting outside the horizon, and C is the partner mode to B inside the horizon. Adapted from [37]

Now, consider a newly emitted mode of late radiation B sitting just outside the horizon. By unitarity late radiation is entangled with early radiation, so B has an entanglement partner in the early radiation, which we call A . Also, since the horizon is smooth

²⁶This is not even mentioned and taken as being implicitly true in [4].

spacetime is locally Minkowski near the horizon and the horizon is locally Rindler as in (2.13). Using the effective field theory description, the near-horizon state is therefore locally given by the Minkowski vacuum and has entanglement across the horizon, as shown with $|\Psi\rangle_M$ in (1.24). Therefore B also has a partner inside the horizon, which we call C . These are shown in Figure 17.

The mode B is then maximally entangled with two different partners. However, this violates monogamy of entanglement since the Hilbert space can be decomposed as above. This is simplest to see by using the strong subadditivity of entropy [4,9,29]

$$S_{AB} + S_{BC} \geq S_B + S_{ABC} \quad (3.2)$$

Old black holes have decreasing entropy so $S_{AB} < S_A$ and when the horizon is smooth $S_{BC} = 0$ so $S_{ABC} = S_A$. Then (3.2) becomes

$$S_A > S_B + S_A$$

implying $S_B < 0$ which fails horribly since the late radiation is thermal with $S_B > 0$.

Thus the AMPS paradox as presented above implies we need to sacrifice unitarity, the equivalence principle, or our effective theory of gravity.²⁷ This is quite disturbing since the first two are *sacred* principles in physics, and we generally have faith in the validity of the effective description, especially considering we could imagine large black holes where the effect of gravity is weak. AMPS took the most conservative resolution to be giving up the smooth horizon. [4]

This is referred to as there being a *firewall* since any observer crossing the horizon will be met by a wall of highly energetic particles. This is the case since we know the spacetime near the horizon is approximately Rindler. The state locally should then be the Minkowski vacuum (1.24), where the entanglement ensures the horizon appears smooth. Upon disrupting the entanglement, as AMPS concludes we should, then the state near the horizon can no longer be the Minkowski vacuum. This leads to a highly excited state and a large energy density at the horizon. [4,15,20]

²⁷or the decomposition.

4. ER = EPR

This chapter discusses $ER = EPR$ and its implications for black hole evaporation, including how $ER = EPR$ “solves” the AMPS paradox.

4.1. The conjecture

We have seen both entanglement and Einstein-Rosen bridges appear non-local. We can find another similarity by considering LOCC.

We imagine two separated systems, A and B . LO refers to any local operations Alice or Bob can perform on their respective systems, while CC refers to classical communication. By this we mean Alice and Bob are only allowed to send *classical* information to each other. For example, Alice may perform measurements on her system and communicate the outcomes to Bob. However, LOCC prohibits Alice from sending a qubit²⁸ directly to Bob. LOCC is importantly unable to create or increase entanglement between A and B . We can only entangle A and B by bringing the systems together or by using a resource of already entangled states. [5,9,38]

Now, consider two separated black holes with no Einstein-Rosen bridge. Creating an Einstein-Rosen bridge between these is impossible. However, we can create a pair of black holes with an Einstein-Rosen bridge and then separate them. We can also create an Einstein-Rosen bridge between two separated black holes by using another pair of black holes with an Einstein-Rosen bridge. This is done by merging members of each pair. This is completely analogous to LOCC. [5]

We could even imagine creating many pairs of small black holes with Einstein-Rosen bridges. We give half of each pair to Alice and the other to Bob. Now, we let Alice and Bob separate after which they merge all their small black holes. We will be left with a pair of black holes with an Einstein-Rosen bridge.

We can push this experiment further by creating many pairs of entangled particles. We give half of each pair to Alice and the other to Bob. Now, we let Alice and Bob

²⁸or any quantum system

separate after which they collapse their share of particles. We will be left with a pair of entangled black holes. Susskind and Maldacena claim these will also have an Einstein-Rosen bridge. [5]

This is generalised by them claiming even a single pair of entangled particles will be connected through some highly quantum Einstein-Rosen bridge. This proposed equivalence between entanglement and Einstein-Rosen bridges is summarised by the symbolic equation [5]

$$\text{ER} = \text{EPR}$$

which is admittedly a wild conjecture. However, we already saw such an equivalence occur in the case of a Schwarzschild black hole in anti-de Sitter spacetime.

4.2. The emitted radiation

To see some implications of $\text{ER} = \text{EPR}$ consider an evaporating black hole. The emitted radiation is entangled with the black hole. So by $\text{ER} = \text{EPR}$ each quanta of radiation will be connected with the black hole through an Einstein-Rosen bridge. We can illustrate this as in Figure 18.

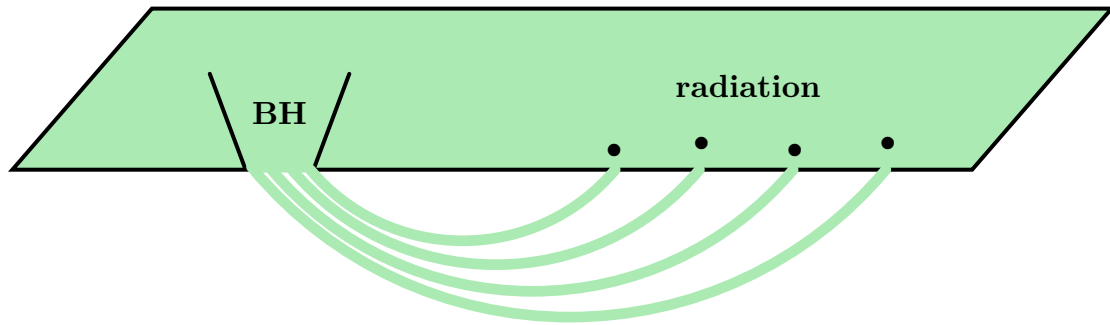
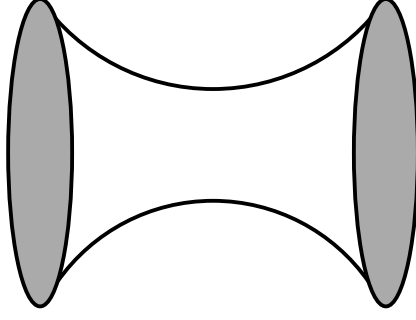
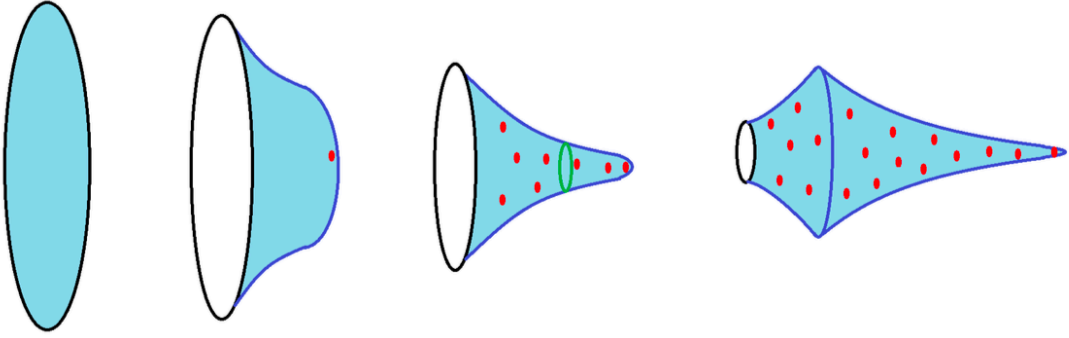


Figure 18: The Einstein-Rosen bridge between a black hole and its emitted radiation.

We can also try to illustrate the evolution of the Einstein-Rosen bridge over time, assuming it can be described geometrically, with embedding diagrams as in Figure 19b. The familiar embedding diagram for a typical Einstein-Rosen bridge is also shown in Figure 19a.



(a) The embedding diagram for an Einstein-Rosen bridge between exteriors L and R in the fully-extended Schwarzschild geometry.



(b) The evolution of the Einstein-Rosen bridge between a black hole and its emitted radiation [5].

Figure 19: Embedding diagrams.

The far left shows a newly formed black hole with the black circle representing the horizon as seen from the outside. As the black hole evaporates, it emits radiation. The red dots show where quanta of radiation are connected to the main body of the Einstein-Rosen bridge. The green circle represents a slice through the Einstein-Rosen bridge, dividing the system into two parts. Radiation to the right of the slice was emitted earlier than radiation to the left. So we can then think of these slices as being proxies of time. [5]

The entanglement entropy of the radiation S_R should as follow the Page curve as to avoid violations of unitarity. Now, we claim $S_R(\text{a slice}) \propto A_{\text{slice}}$ [5].²⁹ This kind of area law relating entanglement entropy to geometry was originally motivated by the Bekenstein-Hawking entropy (2.16) where $S_{\text{BH}} \propto A$. Early computations by Bombelli

²⁹see also [39].

et al. [40] and Srednicki [41] explicitly showed the entanglement entropy of quantum fields obeys

$$S_{\text{EE}} \propto \frac{A}{\epsilon^2}$$

where ϵ is a UV-cutoff length scale. Identifying the cutoff with the Planck length ℓ_p gives

$$S_{\text{EE}} \propto \frac{A}{\ell_p^2}$$

which is S_{BH} up to a numerical factor making the analogy with black hole entropy manifest.

We know S_R is maximal at the Page time, which corresponds to the blue circle in Figure 19b. The Einstein-Rosen bridge at later times should then begin shrinking in accordance with the Page curve since $S_R \propto A$. As more and more radiation is emitted the shape eventually develops into the narrow, horn-like structure shown [5]. With this view the Einstein-Rosen bridge can be seen as a geometric Page curve. However, we stress Figure 19b is speculative and assumes the Einstein-Rosen bridge even has a geometric description.

4.3. Smoothness of the horizon

Consider an observer Bob sitting outside an old black hole. Another observer Alice, of which Bob is unaware, has been collecting the emitted radiation. When she has more than half the radiation under control we imagine she collapses it to form a black hole. This newly formed black hole will be entangled with Bob's black hole. Now, Alice can in principle operate on her black hole and form the Hartle-Hawking state (2.14).³⁰ [5]

With the above setup we will now consider an operational version of the AMPS paradox. This is done since reframing the paradox allows us to see what dynamically changes upon imposing $\text{ER} = \text{EPR}$.

Consider a mode B which is about to be emitted from Bob's black hole. Alice has at some earlier time distilled a mode from her black hole, namely the mode that B will

³⁰ignoring limits on computation.

be entangled with. We denote this mode by A as to be consistent with our previous notation. By distill we mean the operation

$$\text{early radiation} \xrightarrow{\text{distill}} A \otimes (\text{everything else})$$

so Alice separates A from her black hole. Such an operation is in principle possible assuming evaporation is unitary and all information is preserved.

With A in her possession Alice flies over to Bob's black hole in time to meet B when it is emitted. Then following our previous logic B and A are entangled implying B cannot be entangled with the usual partner mode C behind the horizon. The disruption of the BC entanglement means Alice will encounter a particle upon crossing the horizon. [5]

So whenever Alice performs the above experiment for some B she will find the corresponding BC entanglement to be disrupted. We refer to this as an operational version of the argument since the problem with monogamy only becomes apparent once Alice brings A to B . Now AMPS assumes Alice is superfluous. We take this to mean the BC entanglement is disrupted even when she does not perform the experiment. Upon performing the experiment she merely reveals the disruption which was already present. The implication is that all possible BC pairs are disrupted initially creating a firewall. This assumption is quite reasonable since nothing Alice does should lead to the creation of the particle behind the horizon. This stems from AMPS assuming we can decompose the Hilbert space as in (3.1).

By $\text{ER} = \text{EPR}$ this assumption is false. The particle Alice observes upon crossing the horizon was simply created as the other end of the Einstein-Rosen bridge when she distilled A . [5] Thereby removing the necessity for more than a single BC pair being disrupted. Clearly this would not be a viable explanation without imposing $\text{ER} = \text{EPR}$ since there would be no such Einstein-Rosen bridge.

We have actually shown $\text{ER} = \text{EPR}$ implies something *stronger* about what happens to Bob when he decides to cross the horizon.³¹ We can imagine Alice performing various

³¹or weaker depending on ones interpretation of stronger.

operations on her black hole or the radiation. Bob will then as argued be affected by these. She could in principle arrange for Bob to be met by a shockwave or bomb such that he would experience a firewall upon crossing the horizon. She could also do nothing, in which case the horizon would be smooth. Now, her actions are in principle determined by the initial state of the system. Thus what Bob experiences is probabilistic in the quantum sense. We conclude he may be met by a firewall, a smooth horizon, or some bumpy middle ground. One could argue the odds of him meeting a firewall are actually pretty slim since this requires many BC pairs being disrupted. Nonetheless, we can certainly conclude the presence of a firewall is uncertain. [5]

5. Discussion

We know large black holes in anti-de Sitter spacetime have

$$|\text{TFD}\rangle \Leftrightarrow \text{ER-bridge}$$

where $|\text{TFD}\rangle = |\Psi\rangle_{\text{HH}}$, note the $\text{ER} = \text{EPR}$ connection. This is physically meaningful unlike eternal black holes in Minkowski which are also described by $|\Psi\rangle_{\text{HH}}$, however, these are unphysical.

what is certain?

$\text{ER} = \text{EPR}$ is a wild idea, but as we have hopefully shown it is a well motivated idea and similar connections are known to appear in physics. However, $\text{ER} = \text{EPR}$ is also a very vague idea and thus currently the usefulness is questionable. Likewise, $\text{ER} = \text{EPR}$ is hard to disprove.

The AMPS paradox and the information paradox more broadly is a problem which as motivated many ideas in physics. We have seen $\text{ER} = \text{EPR}$ “solves” the AMPS paradox by offering an explicit reason as to why the naive decomposition AMPS uses fails. Other modern ideas such as the island formula offer a way to compute the Page curve, thereby solving the information paradox, without assuming the decomposition AMPS does. The island formula is especially interesting in the context of $\text{ER} = \text{EPR}$ since it is predicated on the idea that entanglement changes the shape of spacetime and uses replica wormholes. [30,31,37,42] Others simply dismiss the information paradox and the AMPS paradox as being solved by corrections and dismissal of the decomposition AMPS uses.

Another central issue, which is somewhat related, is how path integrals, such as those we used, even give the correct results. We are able to derive things such as entropy without doing any counting of states as one typically would. Path integrals appear to have some knowledge of gravity that we do not impose. This makes everything more obscure as we need to interpret these answers that appear correct in a way that makes sense. [30,31]

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